

Exploring the role of transhumance drove roads on the conservation of ant diversity in Mediterranean agroecosystems

Violeta Hevia · Francisco M. Azcárate · Elisa Oteros-Rozas · José A. González

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Abstract Drove roads are a major feature of Mediterranean countries, where this livestock management system has been practiced for centuries. In Spain, many drove roads have become completely or partially abandoned by herders, and transformed for other land uses. Yet, some major drove roads continue to be used for the passage of livestock, and might exert important effects on the conservation of biodiversity and ecosystem functions, particularly in highly transformed agricultural landscapes. In this study, we compare ant taxonomic and functional diversity on a drove road that is still used by transhumant livestock (the Conquense Drove Road) versus an abandoned road (the Murciana Drove Road). Ant species richness per trap and ant richness per sample unit were significantly higher on the used drove road compared to the abandoned drove road. The used drove road also had a positive edge effect on ant species diversity in adjacent croplands (both herbaceous crops and vineyards). Ant functional diversity was also higher on the used drove road. These results draw attention to the role of drove roads as ecologically unique systems and reservoirs of biodiversity, particularly within intensive agricultural landscapes. These effects, however, are largely dependent on maintaining livestock use.

Keywords Ants · Biodiversity · Functional diversity · Transhumant pastoralism

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V. Hevia · E. Oteros-Rozas · J. A. González (✉)
Social-Ecological Systems Laboratory, Department of Ecology, Universidad Autónoma de Madrid,
Darwin 2, 28049 Madrid, Spain
e-mail: jose.gonzalez@uam.es

F. M. Azcárate
Terrestrial Ecology Group, Department of Ecology, Universidad Autónoma de Madrid, Darwin 2,
28049 Madrid, Spain

Introduction

Traditional management practices have been widely recognized for their role in the conservation of biodiversity in agricultural landscapes, both in academia and in the political arena (Fischer et al. 2012; Oppermann et al. 2012; Wright et al. 2012). Distinctive characteristics of traditionally managed agroecosystems that are associated with higher levels of biodiversity include: substantial amounts of natural or seminatural vegetation and high heterogeneity in land cover at a fine spatial grain (Plieninger et al. 2006), low-input technology under local ecological knowledge systems, and taking full advantage of natural ecological processes (Altieri et al. 1987; Perrings et al. 2006). The emerging concept of “High Nature Value farming” in Europe has developed from the growing recognition that the conservation of biodiversity is highly dependent on the continuation of traditionally managed, low-intensity farming systems (Oppermann et al. 2012). This is particularly true in the Mediterranean Basin, which is a widely recognized biodiversity hotspot (Myers et al. 2000), where traditional management practices are responsible for shaping biologically rich “cultural” landscapes as a result of many centuries of dynamic interaction between people and their natural environment (Blondel 2006).

Transhumance represents one such traditional management practice that was developed by ancient Mediterranean societies to cope with a highly seasonal climate, allowing shepherds to balance the herd’s need for forage material with seasonal peaks in pasture availability (Manzano-Baena and Casas 2010). Transhumance involves the seasonal movement of livestock between upland areas (or higher latitudes) and lowland areas (or lower latitudes), where the livestock occupy summer or winter pastures, respectively (Ruiz and Ruiz 1986; Oteros-Rozas et al. 2012a).

Transhumant livestock movements are a widespread phenomenon worldwide (Olea and Mateo-Tomás 2009), and usually follow traditional routes called drove roads (or drover roads or droveways). Drove roads exist in many countries; however, Spain constitutes a unique case, as the traditional network of drove roads has been in use for centuries and was granted legal protection in 1995 (Drove Roads Act). The network is formed by a mixture of Royal Drove Roads (*Cañadas Reales*), with a legal width of ca. 75 m, and smaller trails, known as *cordeles* (ca. 37 m wide) and *veredas* (ca. 20 m). The entire network covers nearly 1 % of the country, spreading across more than 125,000 km, and covering 421,000 ha in total (Merino and Alier 2004).

Transhumant pastoralism in Spain is currently in decline, being generally perceived as a relict activity (Oteros-Rozas et al. 2012b). Nevertheless, the Spanish government, environmental organizations, and academics have widely recognized the historical, cultural, scenic, and ecological value of transhumance (e.g., Bunce et al. 2004; MARM 2011). For instance, drove roads have been reported to serve as ecological corridors, facilitating connectivity among plant and animal populations (Gómez-Sal and Lorente 2004); thus, drove roads contribute towards maintaining genetic diversity, which would otherwise be diminished by general habitat fragmentation (Fahrig 2003). Livestock that moves along drove roads also serve as a vector for the seed dispersal of many plant species, enriching the biodiversity of connected territories (Manzano and Malo 2006).

Most landscapes that are crossed by major drove roads in Spain are characterized by intensive agricultural practices. Agricultural intensification has been considered as one of the major causes of biodiversity decline (Tilman et al. 2002). Although the magnitude of the effect of intensification might be dependent on farmland features

and management options, there is no doubt that, as agriculture becomes more intensive, biodiversity decreases (Krebs et al. 1999). As a result, agri-environment schemes have been promoted by the Common Agricultural Policy (CAP) of the European Union. These schemes aim to alleviate the ecological impact of modern, intensive agriculture, through rewarding farmers who foster biodiversity by adopting environmentally friendly ways of managing the land (i.e., by enhancing landscape multifunctionality). However, to improve the effectiveness of agri-environment schemes and develop effective monitoring systems, it is crucial to identify management practices and landscape configurations that are more beneficial to biodiversity. In this context, drove roads deserve attention. In the case of Spain, drove roads consist of linear strips of natural grassland vegetation crossing a highly transformed agricultural matrix; hence, drove roads might enhance resilience by creating a mosaic of patches with varied functions, habitat heterogeneity, and landscape diversity (Coughenour 2007). In parallel, drove roads act as a source of important ecosystem services (Oteros-Rozas et al. 2012a), and as reservoirs of biodiversity (Bunce et al. 2004; Azcárate et al. 2012).

Ants have been proposed as a good indicator group of changes in ecosystems, since they are widespread, moderately diverse, functionally significant, and easy to sample (Alonso and Agosti 2000). Furthermore, ants are relatively sedentary, and hence sensitive to changes in the environment that arise at small scales in space and time (Bestelmeyer and Wiens 1996). Ants play an important role in ecological flows, and participate in a wide range of ecological interactions (Hölldobler and Wilson 1990; Alonso and Agosti 2000; Franklin 2012). In addition, ant responses to environmental stress and disturbance are relatively well-understood compared to other invertebrates (Hoffmann and Andersen 2003). Moreover, they are sensitive to changes in soil condition within agricultural systems (Peck et al. 1998). Thus, ants may be a useful tool when monitoring changes in fauna related to agricultural practices (Andersen and Majer 2004). Furthermore, ants are an important part of the animal biomass, and act as engineer species (Folgarait 1998). For instance, recent data show that the presence of ants contributes positively to improve certain physico-chemical properties of soils (including the presence of mineral nitrogen, increased porosity, and water infiltration) in areas with a dry climate and intensive agriculture (Evans et al. 2011).

The relationship between species diversity and ecosystem function is known to be complex and context dependent (Naeem and Wright 2003). Traditionally, species richness has been used as the best way to quantify species diversity (Duelli and Obrist 1998; Kremenka et al. 2011). However, in recent years, many studies have focused on assessing functional diversity (Flynn et al. 2009; Laliberté et al. 2010) because it may be directly linked to ecosystem processes (Reiss et al. 2009).

In this contribution, we explore how the presence of a drove road within an intensive agricultural landscape influences the taxonomic and functional diversity of ants. Specifically, we compare two drove roads surrounded by similar agroecosystems, one still in use by transhumant livestock and one that has been abandoned. Within this framework, we analyze differences in ant species composition, species richness, and functional diversity between the two drove roads and between the drove roads and their adjacent croplands. Finally, we discuss the implications of maintaining livestock use of drove roads for the conservation of biodiversity and ecosystem functions in intensive agricultural landscapes.

Methods

Study area

The study site is a fairly homogeneous area in the Autonomous Community of Castilla-La Mancha (Spain), within the municipalities of Las Pedroñeras, Casa de los Pinos, and Villarrobledo (Fig. 1). The area is an open plain, at an altitude of about 800 m a.s.l., and is characterized by a continental Mediterranean climate (mean annual precipitation: 461 mm; mean annual temperature: 14 °C). The lithology is mainly composed of gravels, conglomerates, sands, and sandstone. The predominant land use types are intensive agricultural systems of cereals, onion, garlic, and vineyards.

Two major drove roads, officially demarcated with boundary stones, cross the agricultural landscape of the study area: the Conquense Drove Road (CDR) and the Murciana Drove Road (MDR) (Fig. 1). The CDR is one of the few major drove roads in Spain that still maintains livestock use on foot every year. Fifteen transhumant herders, with almost 8,900 sheep and 250 cows, cross this drove road twice a year in a 25–30-day trip between the summering areas located in the mountain pasturelands of Montes Universales and Serranía de Cuenca (Teruel, Cuenca and Guadalajara provinces) and the southern *dehesas* of Sierra Morena (Jaén and Ciudad Real provinces), approximately 410 km away (Oteros-Rozas et al. 2012a). In contrast, the MDR is an abandoned branch that connects with pasturelands in the Murcia province, and has not been used by transhumant livestock for more than 20 years (Bacaicoa et al. 1993). The vegetation of the two drove roads is mostly composed of Mediterranean grasslands dominated by therophyte species, although some parts of the MDR have been invaded by neighboring croplands, as a consequence of the lack of livestock use.

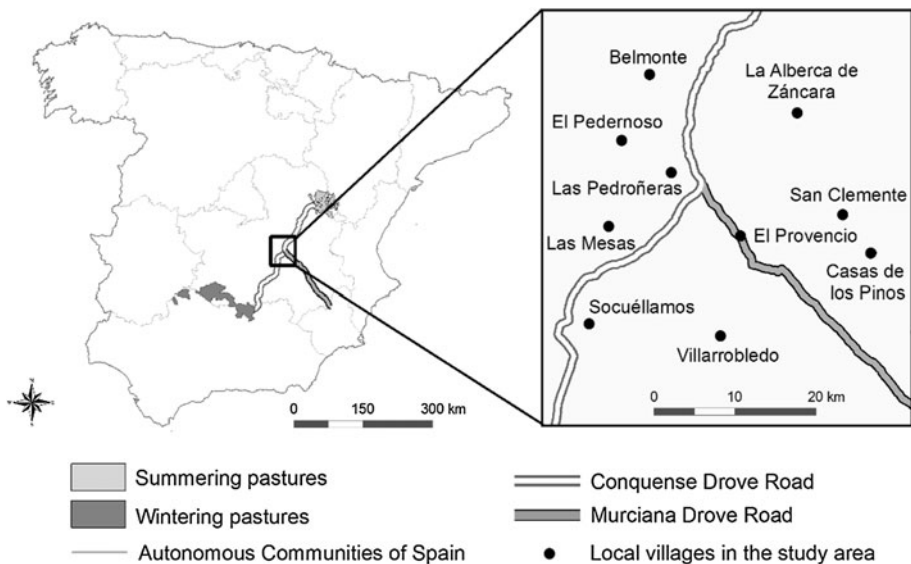


Fig. 1 Study area location, showing the Conquense Drove Road (with its summering and wintering pasturelands) and the Murciana Drove Road

Sampling design and data collection

Two sections of these drove roads, sharing the same biogeophysical characteristics (Supplementary material 1) and situated 14 km apart, were selected for sampling. Sixteen sampling units were positioned on the CDR area and 16 on the MDR area. Since *crop type* may have an important effect on ant community composition (Perfecto et al. 2003; Rosumek et al. 2009), eight sampling units of each drove road were randomly located in environments with *herbaceous crops*, while the other eight sampling units were located in environments with *vineyards*. The distance between sampling units ranged from 291 to 730 m. For each sampling unit, we made one observation *within the drove road* and one observation in the *croplands adjacent to the drove road*. Each observation consisted of a group of six pitfall traps that were placed systematically every 10 m at a 30° angle (Fig. 2). Traps were 2 cm in diameter and 5 cm deep, containing a mixture of 70 % ethanol and 30 % monoethylene glycol (Azcárate and Peco 2011). The traps were placed on 27 July 2011 and collected on 3 August 2011. Despite some limitations, pitfall traps are considered the most objective and fast method to sample ants that live in soil (Andersen 1991; Nash et al. 2004), being recommended for intersite comparisons (Schmidt et al. 2012).

Ants were identified to the species-level with a binocular microscope, and were then characterized according to: (a) head length, (b) functional group, and (c) trophic-based classification. Head length was used as a proxy for overall body size (Kaspari and Weiser 1999), and was measured as the maximum longitudinal length from the most anterior part of the clypeus to the occipital margin, in full face view (Espadaler 2001). We considered functional groups from the perspective of Gitay and Noble (1997); namely, groups of species that respond to disturbance and stress in a similar way. We followed the classification by Roig and Espadaler (2010), who delineated eight functional groups: invasive and/or exotic (IE), generalists and/or opportunistic (GO), social parasites (P), specialist predators (SP), coarse woody debris specialists (CWDS), cold-climate specialists and/or shade habitats (CCS/SH), hot climate specialists and/or open habitats (HCS/OH), and cryptic (C). This classification was adapted from Brown (2000) for the Iberian Peninsula and Balearic Islands. Based on existing literature (Azcárate and Peco 2011) and our field experience at

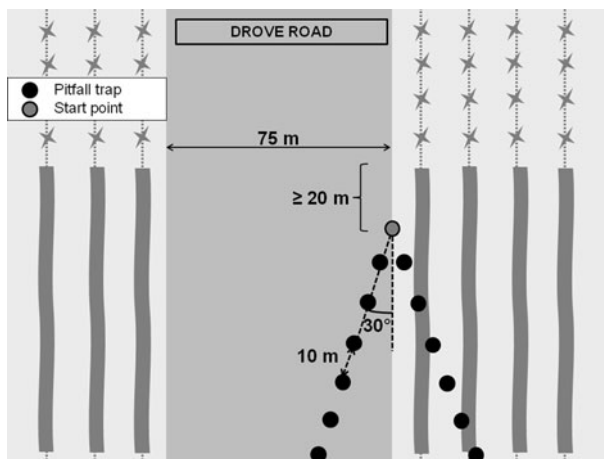


Fig. 2 Schematic representation of the sampling design. This method was carried out at each of the 32 sampling units that were placed in the area of the two drove roads

the study site, we assigned four non-exclusive trophic categories to the ant species; specifically, honeydew, live prey, seeds, and scavenger.

Data analysis

Each observation was characterized by ant species composition, ant richness per sample unit, ant richness per trap, and ant functional diversity. The number of individuals captured in traps was recorded, but not taken into account for the analyses, because it might have been highly influenced by the position of nests (Andersen 1991).

Ant species composition was defined as the number of occurrences (0–6) of each species at each observation. To simplify the composition of species to fewer dimensions, and facilitate interpretation, observations were organized by a non-metric multidimensional scaling based on Euclidean distances and a stress limit of 0.2. The position of observations in each of the two axes was analyzed by a MANOVA test, with two between subject factors (*livestock use* and *crop type*) and one within-subject factor (*drove road*).

To analyze the effect of livestock use and crop type on ant richness per sample unit and ant richness per trap, ANOVA tests were performed with two between-subject factors (*livestock use* and *crop type*) and one within-subject factor (*drove road*). Subsequently, we carried out a post hoc Tukey test to analyze differences between the different levels of the three factors.

Regarding functional diversity, we computed a new matrix of species functional dissimilarities using the Gower distance to calculate functional diversity, with the three traits (head length, functional group, and a trophic-based classification), each with equal weighting (i.e., one third) in the calculations. Ant functional diversity for each observation was calculated using the Rao index of diversity (1), which is widely accepted as an efficient functional diversity index. Because the Rao index of diversity is a generalization of the Simpson's index of diversity, it may be used with various measures of dissimilarity between species, and has an intuitive interpretation (Lepš et al. 2006):

$$FD_{Rao} = \sum_{i=1}^S \sum_{j=1}^S d_{ij} p_i p_j \quad (1)$$

where d_i expresses the dissimilarity between each pair of coexisting species i and j . FD_{Rao} varies from 0 (when two species have the same traits) to 1 (when two species have completely different traits). In the event of binary or categorical traits, $d_{ij} = 0$ when the species have the same trait value, and $d_{ij} = 1$ when the species have different trait values.

Since FD_{Rao} did not fulfill the normality and homoscedasticity requirements for parametric tests, we used a Wilcoxon test to analyze the effect of the drove road factor, and two Mann–Whitney U tests to analyze the effect of livestock use and crop type. Data processing and statistical analyses were performed using the software STATISTICA 8.0.

Results

Ant species richness

Six sampling units (one in the vineyard CDR, three in the herbaceous crop CDR, one in the herbaceous crop MDR, and one in the vineyard MDR) were eliminated, because of some pitfall traps being lost or damaged. For the remaining 26 sampling units, a total of 6,513 ant workers belonging to 26 species were captured (Table 1). Twenty-five species were

recorded at the CDR area (of which 12 were exclusive to this drove road), and 14 at the MDR area (of which 1 was exclusive to the adjacent vineyards).

Ant richness per sample unit was significantly higher on the CDR than on the MDR ($F_{1,28} = 48.486$, $P < 0.001$). Ant richness per sample unit was also significantly higher on the CDR than in the adjacent croplands ($F_{1,28} = 22.207$, $P < 0.001$). We did not detect significant differences for this variable among different crop types ($F_{1,28} = 0.045$, $P = 0.834$). In addition, ant richness per sample unit in herbaceous crops adjacent to the CDR was significantly higher than on the MDR (Tukey's test, $P < 0.01$). However, ant richness per sample unit in vineyard crops adjacent to the CDR was not significantly different to vineyard or herbaceous crops adjacent to the MDR (Tukey's test, $P > 0.05$; Fig. 3). Higher species richness at the CDR area was not only associated with generalist and/or opportunistic species (65 % higher), but particularly with hot climate and/or open habitat specialist species (125 % higher) (Fig. 4).

Table 1 Ant species found on drove roads (DR), along with the functional groups (FG) to which they belong (CCS/SH cold-climate specialists and/or shade habitats, C cryptic, GO generalists and/or opportunistic, HCS/OH hot climate specialists and/or open habitats)

Species	FG	CDR				MDR			
		Herbaceous		Vineyard		Herbaceous		Vineyard	
		DR	Crop	DR	Crop	DR	Crop	DR	Crop
<i>Lasius grandis</i>	CCS/SH	X	X	X	X	X	X	X	X
<i>Temnothorax universitatis</i>	C	X	X		X	X	X		
<i>Solenopsis monticola</i>	C				X				
<i>Aphaenogaster gibbosa</i>	C	X	X	X	X				X
<i>Tapinoma madeirense</i>	GO	X	X	X	X	X	X	X	X
<i>Tapinoma nigerrimum</i>	GO	X	X	X	X	X	X		X
<i>Pheidole pallidula</i>	GO	X	X	X	X	X	X	X	X
<i>Crematogaster auberti</i>	GO	X		X					
<i>Crematogaster sordidula</i>	GO			X					
<i>Cardiocondyla batesii</i>	GO	X		X					
<i>Aphaenogaster iberica</i>	GO	X	X	X	X			X	X
<i>Aphaenogaster senilis</i>	GO	X	X					X	
<i>Tetramorium caespitum</i>	GO		X						X
<i>Tetramorium forte</i>	GO	X	X			X	X		X
<i>Tetramorium semilaeve</i>	GO	X	X						
<i>Cataglyphis rosenhaueri</i>	HCS/OH	X	X	X	X	X	X	X	X
<i>Messor barbarus</i>	HCS/OH	X	X	X	X	X	X	X	X
<i>Messor structor</i>	HCS/OH	X	X						
<i>Cataglyphis iberica</i>	HCS/OH	X	X	X	X	X	X	X	X
<i>Messor bouvieri</i>	HCS/OH	X	X	X					
<i>Messor hispanicus</i>	HCS/OH	X		X					
<i>Oxyopomyrmex saulcyi</i>	HCS/OH								X
<i>Camponotus micans</i>	HCS/OH	X		X					
<i>Camponotus aethiops</i>	HCS/OH	X	X	X	X				
<i>Camponotus sylvaticus</i>	HCS/OH	X							
<i>Camponotus foreli</i>	HCS/OH			X					

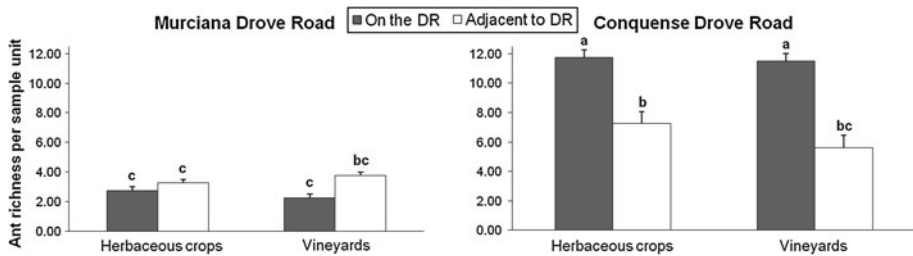


Fig. 3 Ant richness per sample unit on the two drove roads. Error bars represent mean $\pm$ standard error. Different letters indicate significant differences for this variable (Tukey's test, $P < 0.05$) (DR drove road)

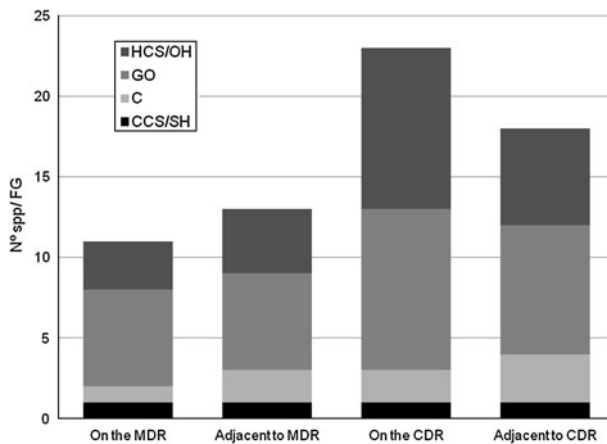


Fig. 4 Number of ant species for each functional group on the two drove roads (CCS/SH cold-climate specialists and/or shade habitats, C cryptic, GO generalist and/or opportunistic, HCS/OH hot climate specialist and/or open habitats, CDR Conquense Drove Road, MDR Murciana Drove Road)

Species richness per trap was higher on the CDR than on the MDR ($F_{1,28} = 27.42$, $P < 0.001$). Richness per trap was also higher on the CDR than in adjacent croplands ($F_{1,28} = 25.02$, $P < 0.001$). Crop type did not yield significant differences ($F_{1,28} = 0.87$, $P = 0.358$), although herbaceous crops adjacent to the CDR showed a significantly higher value compared to the abandoned drove road and its adjacent areas (Tukey's test, $P < 0.05$; Fig. 5).

Ant species composition

Non-metric multidimensional scaling of ant species composition revealed three main groups of observations: (1) the abandoned drove road and its adjacent fields, (2) the inside of the drove road in use, and (3) the croplands adjacent to the drove road in use (stress = 0.148) (Fig. 6). The positioning of the observations in the multidimensional scaling was dependent on livestock use ($F_{2,27} = 81.72$; $P < 0.001$) and the drove road ($F_{2,27} = 44.89$; $P < 0.001$). However, there were no significant differences in relation to crop type ($F_{2,27} = 0.4479$; $P > 0.05$).

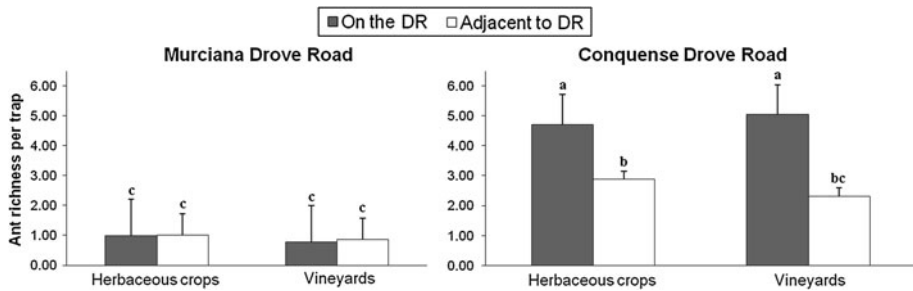


Fig. 5 Ant richness per trap on the two drove roads. Error bars represent mean $\pm$ standard error. Different letters indicate significant differences for this variable (Tukey's test, $P < 0.05$) (DR drove road)

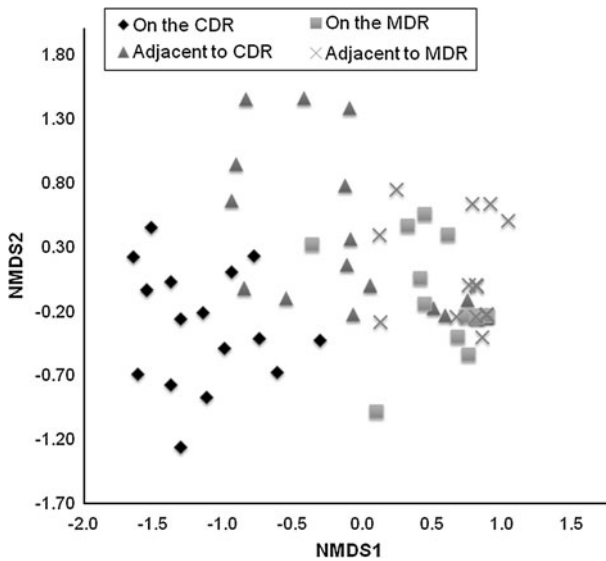


Fig. 6 Non-metric multidimensional scaling (NMDS) of ant species composition on the two drove roads. Dissimilarities were calculated by Euclidean distance (CDR Conquense Drove Road, MDR Murciana Drove Road)

Functional diversity

The Rao index showed significantly higher values on the CDR compared to the MDR ($U = 10$; $P < 0.05$). We also found significant differences between samples on the drove roads and samples in their adjacent croplands; higher values on the CDR compared to adjacent croplands ($Z = -2.741$; $P < 0.05$), but lower values on the MDR compared to adjacent croplands ($Z = -2.353$; $P < 0.05$). Functional diversity did not respond to crop type in any of the compared areas (on the CDR: $U = 31$; $P > 0.05$; croplands adjacent to CDR: $U = 18$; $P > 0.05$; on the MDR: $U = 29$; $P > 0.05$; croplands adjacent to MDR: $U = 30$; $P > 0.05$) (Fig. 7).

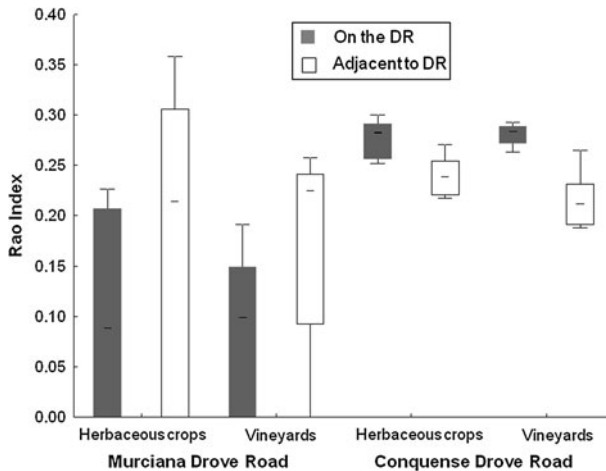


Fig. 7 Rao index values on the two drove roads. The *boxes* represent the three quartiles, and the *whiskers* represent the minimum and maximum values obtained for this variable (*DR* drove road)

Discussion

Ant diversity and drove roads

Our results suggest that the presence of a well-preserved drove road used by livestock might have significant effects on ant assemblages. Ant taxonomic diversity was significantly higher on the used drove road compared to the abandoned drove road. Moreover, the edges of croplands adjacent to the used drove road showed greater ant taxonomic diversity compared to the abandoned drove road and its adjacent croplands. In contrast, ant assemblages on the abandoned drove road were minimally different from those on the edges of adjacent croplands.

The simplest explanation for these differences is that the active drove road habitat is more favorable for ants. For instance, the environment of the used drove road has minimal or no inputs of agrochemicals, but periodical inputs of manure. Yet, the absence of tilling within the drove road reduces disturbance compared to the adjacent cropland matrix. These factors might influence the quantity and variety of food that is available to ants, which might explain the observed differences in species composition and taxonomic diversity between the two roads. Some agricultural practices (e.g., soil fertilization by chemical inputs, plowing, and irrigation) have been related with a reduction in ant diversity and ant colony density (Díaz 1991). These practices are widespread in the study area but are not conducted on drove roads that maintain livestock use (pers. obs.).

Our results also suggest that drove road usage has a positive effect on ant species diversity in adjacent croplands, at least on the edges of these croplands. Some studies have demonstrated that, in warm and dry regions, the presence of ants in croplands facilitates water infiltration, the supply of nitrogen, and the porosity of the soil, which are characteristics associated with the higher production of some cereals (Evans et al. 2011). Therefore, it might be expected that the presence of a used drove road close to croplands would facilitate physical and chemical changes to the soil, which might favor an increase in the productivity. Future research in croplands adjacent to used and abandoned drove

roads would be necessary to test potential effects on productivity-yields and explore if these effects extend beyond the borders of the drove road.

The crop type (vineyard or herbaceous) present in the adjacent matrix of both drove roads did not appear to influence the taxonomic and functional diversity of ants. Changes in vegetation structure that occur in environments with a high percentage of intensive cultivation are considered to be followed by changes in the structure of the faunal community in the given area (Whitford 1997). The lack of differences found in this study might be due to a change in vegetation structure, which has a similar effect on ant populations inhabiting both vineyard and herbaceous crops, causing a reduction of available niches or impacts on the microclimate (Gómez et al. 2003).

Regarding ant functional diversity, our results showed that the FD_{Rao} values in the CDR area were higher compared to the MDR area (although these differences were not as marked as those observed for taxonomic diversity). There were also differences in FD_{Rao} on and adjacent to the drove roads, but not between crops adjacent to the CDR and the MDR. Several empirical studies have demonstrated that functional diversity more often determines ecosystem functioning than species richness (Díaz et al. 2006). In particular, functional traits have been identified as the key mechanism by which single species and groups of species influence ecosystem properties (de Bello et al. 2010). Functional diversity has been used to detect major ecological impacts associated with land use change, with this functional approach being particularly useful in the absence of species-level knowledge (Hausner et al. 2003; Andersen and Majer 2004). Our results suggest that the drove road used by livestock has a higher potential as a functional reservoir compared to the abandoned drove road. This observed difference might have important implications for ecosystem functioning, and the supply of ecosystem services at the landscape level, and also for coping with disturbances of both natural and anthropogenic origin (Lavorel et al. 2007).

However, caution should be taken with the interpretation of our results, as they are heavily location-dependent, and only applicable to the specific conditions of intensive agricultural landscapes in the study area. Under different ecological conditions, drove roads might have different effects on ant diversity, probably related with the characteristics of the surrounding matrix. For example, Azcárate et al. (2013) found that drove roads increase ant diversity in forest landscapes, but exert little or no effect in open environments (rangelands or extensive traditionally managed croplands). Therefore, additional studies about the effect of drove roads on ant diversity at a local scale are required to objectively inform land management at regional or larger spatial scales.

Policy and management implications

Changes in land use are among the main causes of biodiversity loss in Mediterranean ecosystems (Sala et al. 2000). In particular, agriculture intensification has affected the biodiversity of many insect species (Tscharntke et al. 2005); yet, the effects on ant communities remain poorly understood (de Bruyn 1999). Some studies using ants as bioindicators in Australia showed that the main cause of biodiversity loss is agriculture intensification (Majer and Beeston 1996). Moreover, agricultural practices that alter the species composition and/or reduce biodiversity, in agricultural systems and other non-agricultural systems, have a negative impact on ecosystem services (Tilman et al. 2002). This phenomenon arises because the capacity of these systems to provide services is highly dependent on the number of species present (Loreau et al. 2001).

Ants have been widely demonstrated to serve as robust ecological indicators (Andersen and Majer 2004). For instance, the extent of change in ant assemblages may be used as an indicator of broader changes in ecosystems integrity (Andersen et al. 2004). Our study shows that, in a highly disturbed matrix of intensive agricultural fields, a well-preserved drove road with livestock use is associated with higher ant species richness and higher ant functional diversity compared to an abandoned drove road. This observation might have potential implications on ecosystem function and overall biodiversity conservation at a landscape scale. It could be argued that the observed higher species richness is simply a consequence of the presence of opportunistic or common species, which would lower its interest from a species-level conservation perspective. However, it should be noted that, in this case-study, the observed increase was not only due to the presence of generalist/opportunistic species, but also due to the presence of specialist species (see Fig. 4). Although there is no published information available about the status of regional ant fauna, when taking the $1^\circ \times 1^\circ$ grid cells of the Iberian Peninsula as an indicator (www.hormigas.org), we found that seven of the species recorded in the CDR have been recorded in less than 25 % of the cells, for only three of the species found in the MDR. In any case, future research with a species-level approach would be of high interest to fill this knowledge gap and complement our results regarding the conservation relevance of drove roads for ants.

At a larger scale, spatially complex systems (in terms of structural complexity and spatial heterogeneity of vegetation) often tend to exhibit greater species richness (MacArthur 1972). In this case, drove roads are diversifying features that increase habitat heterogeneity at the landscape level, acting as ecological corridors and biodiversity reservoirs (Manzano and Malo 2006; Coughenour 2007; Azcárate et al. 2012), particularly within a matrix of highly transformed, intensive agricultural landscapes. Transhumant livestock movements and the presence of extensive drove road networks are also responsible for the maintenance of critical ecosystem functions and services (González et al. 2012; Carmona et al. 2013). In the context of the uncertainty that accompanies current patterns of global environmental change, traditional practices, such as transhumance, might contribute toward safeguarding multifunctional, cultural landscapes of high natural value (Herzog et al. 2005; Oteros-Rozas et al. 2013). Such practices potentially enhance the adaptive capacity of landscapes to cope with climate variability and increase social-ecological resilience in Mediterranean agroecosystems (Oteros-Rozas et al. 2012b).

Despite the gradual decline of transhumance in recent decades in Mediterranean Spain, opportunities for the recovery of this practice still exist (Fernández-Giménez and Fillat-Estaque 2012; Oteros-Rozas et al. 2013). The presence of a legally protected network of drove roads is the foundation for this potential recovery, and makes Spain a unique case among developed Mediterranean countries. However, this regulation is not always observed by private and/or public users, leading to abuse and misuse, with many drove roads being partially abandoned and absorbed into other land uses; consequently, losing their role as potential biodiversity reservoirs.

Therefore, action is urgently needed that ensures the effective protection of drove roads for their priority function as livestock routes. Moreover, as the maintenance of drove roads is highly dependent on transhumant livestock use, policy measures are also necessary to safeguard the social and economic sustainability of this traditional practice (Herzog et al. 2005; Oteros-Rozas et al. 2013). By dedicating a certain amount of direct payments to “greening”, the new CAP reform aims to ensure that European Union farmers deliver environmental and climatic benefits as part of their daily activities. Therefore, the undergoing CAP reform provides a window of opportunity for the development of mechanisms (e.g., payment for ecosystem services schemes, quality labeling, products

marketing, etc.) to promote customary practices, such as transhumance, that contribute toward reversing declines in farmland biodiversity throughout Europe.

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References

- Alonso LE, Agosti D (2000) Biodiversity studies, monitoring and ants: an overview. In: Agosti, D, Majer JD, Alonso LE, Schultz TRY (eds) *Ants: standard methods for measuring and monitoring biodiversity*. Smithsonian Institution Press, Washington DC, pp 1–8
- Altieri MA, Kat Anderson M, Merrick LC (1987) Peasant agriculture and the conservation of crop and wild plant resources. *J Conserv Biol* 1:49–58
- Andersen AN (1991) Sampling communities of ground-foraging ants: pitfall catches compared with quadrat counts in an Australian tropical savanna. *Aust J Ecol* 16:273–279
- Andersen AN, Majer JD (2004) Ants show the way down under: invertebrates as bioindicators in land management. *Front Ecol Environ* 2(6):291–298
- Andersen AN, Fisher A, Hoffmann BD, Read JL, Richards R (2004) Use of terrestrial invertebrates for biodiversity monitoring in Australian rangelands, with particular reference to ants. *Aust Ecol* 29:87–92
- Azcárate FM, Peco B (2011) Abandonment of Mediterranean grasslands: consequences for ant assemblages. *Insect Conserv Diver* 5:279–288
- Azcárate FM, Robleño I, Seoane J, Manzano P, Peco B (2012) Drove road as local biodiversity reservoirs: effects on landscape pattern and plant communities in a Mediterranean region. *Appl Veg Sci*. doi:10.1111/avsc.12003
- Azcárate FM, Seoane J, Castro S, Peco B (2013) Drove roads: keystone structures that promote ant diversity in Mediterranean forest landscapes. *Acta Oecol* 49:107–115
- Bacaicoa I, Elías JM, Grande J (1993) Cuadernos de la trashumancia n° 8. Albarracín–Cuenca–Molina. ICONA, Madrid
- Bestelmeyer B, Wiens JA (1996) The effects of land use on the structure of ground-foraging ant communities in the Argentine Chaco. *Ecol Appl* 6:1225–1240
- Blondel J (2006) The ‘Design’ of Mediterranean landscapes: a millennial story of humans and ecological systems during the historic period. *Hum Ecol* 34:713–729
- Brown WL (2000) Diversity of ants. In: Agosti D, Majer JD, Alonso LE, Schultz TR (eds) *Ants: standard methods for measuring and monitoring biodiversity*. Smithsonian Institution Press, Washington DC, pp 122–144
- Bunce RGH, Pérez-Soba M, Jongman RHG, Gómez-Sal A, Herzog F, Austad I (2004) Transhumance and biodiversity in European mountains. IALE, Wageningen
- Carmona CP, Azcárate FM, Oteros-Rozas E, Gonzalez JA, Peco B (2013) Assessing the effects of seasonal grazing on holm oak regeneration: implications for the conservation of Mediterranean dehesas. *Biol Conserv* 159:240–247
- Coughenour MB (2007) Causes and consequences of herbivore movement in landscape ecosystems. In: Galvin KA, Reid RS, Behnke RH, Hobbs NT (eds) *Fragmentation in semi-arid and arid landscapes: consequences for human and natural systems*. Springer, Dordrecht, pp 46–91
- de Bello F, Lavorel S, Díaz S, Harrington R, Cornelissen JHC, Bardgett RD, Berg MP, Cipriotti P, Feld CK, Hering D, Martins da Silva P, Potts SG, Sandin L, Sousa JP, Storkey J, Wardle DA, Harrison PA (2010) Towards an assessment of multiple ecosystem processes and services via functional traits. *Biodivers Conserv* 19:2873–2893
- de Bruyn LA (1999) Ants as bioindicators of soil function in rural environments. *Agr Ecosyst Environ* 74:425–441
- Díaz M (1991) Spatial patterns of granivorous ant nest abundance and nest site selection in agricultural landscapes of Central Spain. *Insect Soc* 38:351–363
- Díaz S, Fargione J, Chapin FS, Tilman D (2006) Biodiversity loss threatens human well-being. *PLoS Biol* 4(8):e277. doi:10.1371/journal.pbio.0040277

- Duelli P, Obrist MK (1998) In search of the best correlates for local organismal biodiversity in cultivated areas. *Biodivers Conserv* 7:297–309
- Espadaler X (2001) Formicine ants comply with the size-grain hypothesis. *Funct Ecol* 15:136–139
- Evans TA, Dawes TZ, Ward PR, Lo N (2011) Ants and termites increase crop yield in a dry climate. *Nat Commun* 2:262. doi:[10.1038/ncomms1257](https://doi.org/10.1038/ncomms1257)
- Fahrig L (2003) Effects of habitat fragmentation on biodiversity. *Annu Rev Ecol Evol Syst* 34:487–515
- Fernández-Giménez ME, Fillat-Estaque F (2012) Pyrenean pastoralists' ecological knowledge: documentation and application to natural resource management and adaptation. *Hum Ecol* 40(2):287–300. doi:[10.1007/s10745-012-9463-x](https://doi.org/10.1007/s10745-012-9463-x)
- Fischer J, Hartel T, Kuemmerle T (2012) Conservation policy in traditional farming landscapes. *Conserv Lett*. doi:[10.1111/j.1755-263X.2012.00227.x](https://doi.org/10.1111/j.1755-263X.2012.00227.x)
- Flynn DFB, Gogol-Prokurat M, Nogeire T, Molinari N, Richers BT, Lin BB, Simpson N, Mayfield MM, DeClerck F (2009) Loss of functional diversity under land use intensification across multiple taxa. *Ecol Lett* 12:22–33
- Folgarait PJ (1998) Ant biodiversity and its relationship to ecosystem functioning: a review. *Biodivers Conserv* 7:1221–1244
- Franklin K (2012) The remarkable resilience of ant assemblages following major vegetation change in an arid ecosystem. *Biol Conserv* 148:96–105
- Gitay H, Noble IR (1997) What are functional types and how should we seek them? In: Smith TM, Shugart HH, Woodward FI (eds) *Plant functional types their relevance to ecosystem properties and global change*. Cambridge University Press, Cambridge, pp 3–19
- Gómez C, Casellas D, Oliveras J, Bas JM (2003) Structure of ground-foraging ant assemblages in relation to land-use change in the northwestern Mediterranean region. *Biodivers Conserv* 12:2135–2146
- Gómez-Sal A, Lorente I (2004) The present status and ecological consequences of transhumance in Spain. In: Bunce RGH, Pérez-Soba M, Jongman RHG, Gómez-Sal A, Herzog F, Austad I (eds) *Transhumance and biodiversity in European Mountains*. Alterra, Wageningen, pp 233–248
- González JA, Oteros-Rozas E, Martín-López B, López CA, Zorrilla P, Montes C (2012) La trashumancia en la Cañada real conquense: valores ecológicos, sociales y económicos asociados a una práctica ganadera tradicional. Universidad Autónoma de Madrid, Madrid
- Hausner VH, Yoccoz NG, Ims RA (2003) Selecting indicator traits for monitoring land use impacts: birds in northern coastal birch forests. *Ecol Appl* 13:999–1012
- Herzog F, Bunce RGH, Pérez-Soba M, Jongman RHG, Gómez-Sal A, Austad I (2005) Policy options to support transhumance and biodiversity in European mountains. *Mt Res Dev* 25:82–84
- Hoffmann BD, Andersen AN (2003) Responses of ants to disturbance in Australia, with particular reference to functional groups. *Austral Ecol* 28:444–464
- Hölldobler B, Wilson EO (1990) *The ants*. Belknap Press, Cambridge
- Kaspari M, Weiser MD (1999) The size-grain hypothesis and interspecific scaling in ants. *Funct Ecol* 13:530–538
- Krebs JR, Wilson JD, Bradbury RB, Siriwardena GM (1999) The second silent spring? *Nature* 400:611–612
- Krewenka KM, Holzschuh A, Tscharnkte T, Dormann C (2011) Landscape elements as potential barriers and corridors for bees, wasps and parasitoids. *Biol Conserv* 144:1816–1825
- Laliberté E, Wells JA, DeClerck F, Metcalfe DJ, Catterall CP, Queiroz C, Aubin I, Bonser SP, Ding Y, Fraterrigo JM, McNamara S, Morgan JW, Sánchez-Merlos D, Vesk PA, Mayfield MM (2010) Land-use intensification reduces functional redundancy and response diversity in plant communities. *Ecol Lett* 13:76–86
- Lavorel S, Grigulis K, McIntyre S, Williams NSG, Garden D, Dorrough J, Berman S, Quéfier F, Thébaud A, Bonis A (2007) Assessing functional diversity in the field—methodology matters! *Funct Ecol* 22(1):134–147
- Lepš J, de Bello F, Lavorel S, Berman S (2006) Quantifying and interpreting functional diversity of natural communities: practical considerations matter. *Preslia* 78(4):481–501
- Loreau MS, Naeem S, Inchausti P, Bengtsson J, Grime JP, Hector A, Hooper DU, Huston MA, Raffaelli D, Schmid B, Tilman D, Wardle DA (2001) Biodiversity and ecosystem functioning: current knowledge and future challenges. *Science* 294:804–808
- MacArthur RA (1972) *Geographical ecology*. Harper Row, New York
- Majer JD, Beeston G (1996) The biodiversity integrity index: an illustration using ants in Western Australia. *Conserv Biol* 10:65–73
- Manzano P, Malo JE (2006) Extreme long-distance seed dispersal via sheep. *Front Ecol Environ* 4:244–248
- Manzano-Baena P, Casas R (2010) Past, present and future of *Trashumancia* in Spain: nomadism in a developed country. *Pastoralism* 1(1):74–90

- MARM (2011) Libro blanco de la Trashumancia en España. Ministerio de Medio Ambiente, Medio Rural y Marino, Madrid.
- Merino J, Alier JL (2004) La multifuncionalidad de las vías pecuarias españolas en el marco del desarrollo rural. *Tecnología y Desarrollo*.
- Myers N, Mittermeier RA, Mittermeier CG, da Fonseca GAB, Kent J (2000) Biodiversity hotspots for conservation priorities. *Nature* 403:853–858
- Naeem S, Wright JP (2003) Disentangling biodiversity effects on ecosystem functioning: deriving solutions to a seemingly insurmountable problem. *Ecol Lett* 6:567–579
- Nash MS, Bradford DF, Franson SE, Neale AC, Whitford WG, Heggem DT (2004) Livestock grazing effects on ant communities in the eastern Mojave Desert, USA. *Ecol Indic* 4:199–213
- Olea PP, Mateo-Tomás P (2009) The role of traditional farming practices in ecosystem conservation: the case of transhumance and vultures. *Biol Conserv* 142:1844–1853
- Oppermann R, Beaufoy G, Jones G (2012) High nature value farming in Europe. Verlag Regionalkultur, Ubstadt-Weiher
- Oteros-Rozas E, González JA, Marín-López B, López CA, Zorrilla-Miras P, Montes C (2012a) Evaluating ecosystem services in transhumance cultural landscapes: an interdisciplinary methodological framework. *GAIA* 21:235–243
- Oteros-Rozas E, González JA, Martín-López B, López CA, Montes C (2012b) Ecosystem services and social-ecological resilience in transhumance cultural landscapes: learning from the past, looking for a future. In: Plieninger T, Bieling C (eds) *Resilience and the cultural landscape: understanding and managing change in human-shaped environments*. Cambridge University Press, New York, pp 185–193
- Oteros-Rozas E, Martín-López B, López CA, Palomo I, González JA (2013) Envisioning the future of transhumant pastoralism through participatory scenario planning: a case study in Spain. *Rangeland J*. doi:10.1071/RJ12092
- Peck SI, Mcquaid B, Campbell CL (1998) Using ant species (Hymenoptera: Formicidae) as a biological indicator of agroecosystem condition. *Environ Entomol* 27(5):1102–1110
- Perfecto I, Mas A, Dietsch T, Vandermeer J (2003) Conservation of biodiversity in coffee agroecosystems: a tri-taxa comparison in southern Mexico. *Biodivers Conserv* 12:1239–1252
- Perrings C, Jackson L, Bawa K, Brussaard L, Brush S, Gavin T, Papa R, Pascual U, De Ruiter P (2006) Biodiversity in agricultural landscapes: saving natural capital without losing interest. *Conserv Biol* 20:263–264
- Plieninger T, Hoecht F, Spek T (2006) Traditional land-use and nature conservation in European rural landscapes. *Environ Sci Policy* 9:317–321
- Reiss J, Bridle JR, Montoya JM, Woodward G (2009) Emerging horizons in biodiversity and ecosystem functioning research. *Trends Ecol Evol* 24:505–514
- Roig X, Espadaler X (2010) Propuesta de grupos funcionales de hormigas para la Península Ibérica y Baleares, y su uso como bioindicadores. *Iberomyrmex* 2:28–29
- Rosumek FB, Silveira FAO, Neves FS, Barbosa NPU, Diniz L, Oki Y, Pezzini F, Fernandes GW, Cornelissen T (2009) Ants on plants: a meta-analysis of the role of ants as plant biotic defenses. *Oecologia* 160:537–549
- Ruiz M, Ruiz JP (1986) Ecological history of transhumance in Spain. *Biol Conserv* 37(1):73–86
- Sala O, Chapin FS, Armesto J, Berlow E, Bloomfield J, Dirzo R (2000) Global biodiversity scenarios for the year 2100. *Science* 287:1770–1774
- Schmidt AC, Fraser LH, Carlyle CN, Basset ERL (2012) Does cattle grazing affect ant abundance and diversity in temperate grasslands? *Rangeland Ecol Manag* 65(3):292–298
- Tilman D, Cassman KG, Matson PA, Naylor R, Polasky S (2002) Agricultural sustainability and intensive production practices. *Nature* 418:671–677
- Tscharntke T, Klein AM, Kruess A, Steffan-Dewenter I, Thies C (2005) Landscape perspectives on agricultural intensification and biodiversity-ecosystem service management. *Ecol Lett* 8:857–874
- Whitford WG (1997) Desertification and animal biodiversity in the desert grasslands of North America. *J Arid Environ* 37:709–720
- Wright HL, Lake IR, Dolman PM (2012) Agriculture: a key element for conservation in the developing world. *Conserv Lett* 5:11–19